

A 14-nm Ultra-Low Jitter Fractional- N PLL Using a DTC Range Reduction Technique and a Reconfigurable Dual-Core VCO

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Abstract—This work presents a 6-GHz low-jitter and high figure-of-merit (FoM) fractional- N phase-locked loop (PLL). It uses a digital-to-time converter (DTC)-based sampling PLL architecture. To achieve ultra-low jitter in fractional- N mode, a phase detector range reduction technique is used to halve the required DTC delay range (DR), resulting in lower thermal noise and better DTC linearity. Moreover, a reconfigurable dual-core voltage-controlled oscillator (VCO) provides extra freedom in power and jitter tradeoff. It achieves 83.4-fs rms jitter in fractional- N mode, integrated from 10 kHz to 100 MHz, with a 76.8-MHz crystal oscillator (XO) reference. In the low-power mode, the rms jitter degrades to 96.3 fs and the PLL FoM improves from -250.1 to -251.2 dB, as the PLL power consumption reduces from 14.2 to 8.2 mW. The measured fractional spurs are less than -70 dBc for near-integer channels. The PLL rms jitter remains within 100 fs across the 5–7-GHz output frequency band, thanks to the digital background calibrations. It is implemented in a 14-nm FINFET process and occupies 0.31 mm².

Index Terms—Delta-sigma modulator (DSM), digital-to-time converter (DTC), fractional- N phase-locked loop (PLL), frequency synthesis, low jitter, multi-modulus divider (MMD), sampling phase detector (SPD), voltage-controlled oscillator (VCO).

I. INTRODUCTION

THE ever-growing demand on higher data rates and low latency has driven the latest and emerging wireless communication protocols to deploy higher modulation scheme, multi-input–multi-output (MIMO) transceiver configuration, and wider aggregated channel bandwidth (BW) to deliver tens of gigabits per second (Gb/s) throughput. The 5G New Radio (NR) uses both the sub-6 GHz and mm-wave frequency bands, i.e., 24.25–52.6 GHz, with up to 256-quadrature amplitude modulation (QAM) to deliver 10 Gb/s and beyond [1], [2]. The next-generation wireless local area network standard IEEE 802.11be-Extremely High Throughput, also known as Wi-Fi 7,

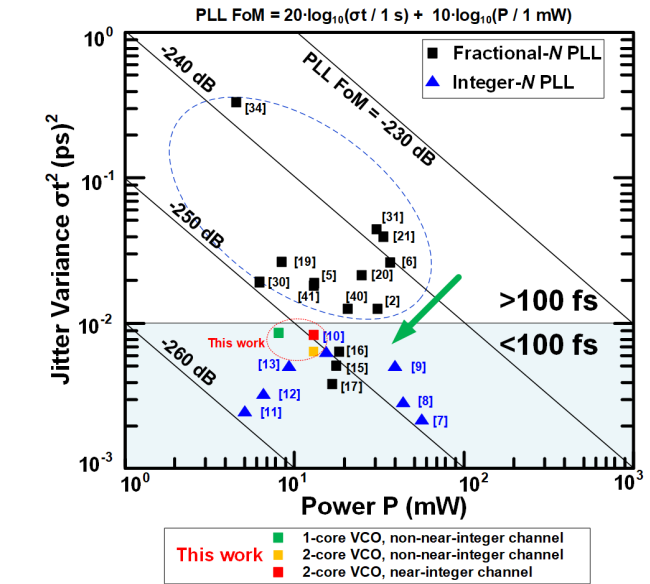


Fig. 1. Jitter variance versus power for state-of-the-art low-jitter PLLs.

aims to support up to 30-Gb/s throughput using 4096-QAM and MIMO enhancement in the 2.4-, 5-, and 6-GHz frequency bands [3]. These emerging communication standards demand ultra-low jitter fractional- N phase-locked loops (PLLs) to generate local oscillator (LO) with sub-100 fs rms jitter to meet the stringent error vector magnitude (EVM) requirement [3], [4]. This is much more demanding than that is required for 4G LTE and Wi-Fi 802.11ax standard specifications (i.e., 150 fs_{rms}) [5], [6].

Recently, numerous works on integer- N PLLs using a sub-sampling phase detector (SSPD) have demonstrated integrated jitter of less than 100 fs or even below 50 fs, with a jitter power figure-of-merit (FoM) as low as -256 dB [7]–[13], as shown in Fig. 1. The SSPD offers very high gain so that the noise contribution from following circuit stages is largely suppressed (e.g., charge pump and loop filter) [14]. As a contrary, only a few fractional- N PLLs reported to date have achieved sub-100 fs_{rms} jitter (see Fig. 1), and their jitter performance at near-integer channels were not reported in [15]–[17], which could increase significantly due to in-band fractional spurs.

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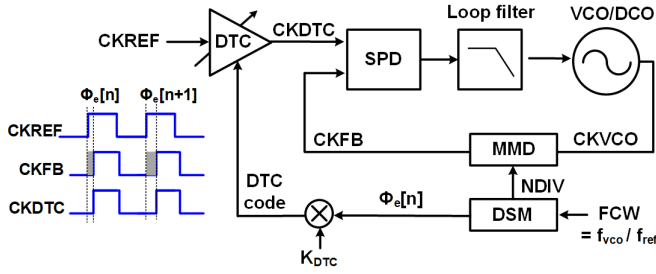


Fig. 2. Block diagram of a DTC-based sampling fractional- N PLL.

Compared with its integer- N counterpart, the major design challenge of a fractional- N PLL lies in the phase detection path. A typical phase error between reference clock (CKREF) and feedback clock (CKFB) in steady-state is up to a few nano-seconds (versus a few pico-seconds at integer- N mode) due to the accumulated phase quantization error (QE), introduced by the delta-sigma modulator (DSM) for fractional- N synthesis [18]. This hinders the use of a high-gain sampling or SSPD in a fractional- N PLL due to its small linear operation range (i.e., a few ps). Recent works in [16], [17], and [19]–[21] used a digital-to-time converter (DTC) in reference clock path to compensate the QE so that a near-zero phase error is presented at the phase detector input after locking (similar to an integer- N case). Fig. 2 shows the block diagram of a DTC-based sampling PLL, which has demonstrated lower jitter and better FoM than other state-of-the-art fractional- N PLLs using charge pumps or time-to-digital converters (TDCs) for phase detection [2], [5]. The sampling phase detector (SPD) in Fig. 2 can be implemented in either an analog form as in [16] and [17] or in a digital form as in [19]. However, there are still some performance gaps between the fractional- and integer- N modes in both jitter and FoM (see Fig. 1), due to the extra random jitter, quantization noise (QN), and power consumption introduced by DTC [20]–[22]. Moreover, the nonlinearity of DTC circuit results in fractional spurious tones and noise-folding, contributing significant amount of integrated jitter, especially at near-integer channels [20], [23]. Thus, DTC remains the bottleneck in such a PLL architecture, as it needs to provide a wide linear delay range (DR) to cancel the QE due to DSM.

To alleviate the aforementioned challenge, this work uses a modified multi-modulus divider (MMD) and the DSM that controls it to reduce the accumulated phase QE by half for any given order of DSM [24], [34]. This technique is demonstrated in a DTC-based analog sampling PLL to reduce DTC thermal noise and improve its linearity, resulting in lower in-band phase noise (PN) and lower fractional spurs. Moreover, several background digital calibrations are used in this design to ensure the ultra-low jitter performance over process, voltage, and temperature (PVT) variations. These calibrations include DTC gain, K_{DTC} , calibration, reference clock duty-cycle correction (DCC), and the voltage-controlled oscillator (VCO) DCC. In addition, low PN oscillator is always needed for low-jitter PLLs, despite the PLL architecture. There, however, is a

lower bound on the size of a planar inductor below which significant quality factor (Q) degradation occurs, limiting VCO PN [25]–[27]. To mitigate this issue, a dual-core VCO is used in this design, which improves VCO PN by 3 dB at the cost of doubled current consumption. To support a low-power mode, the VCO can be reconfigured to a single core to trade jitter for power consumption. Using the above techniques, the 5–7-GHz fractional- N PLL achieves rms jitter as low as 83.6 fs_{rms} (integrated from 10 kHz to 100 MHz) with a 76.8-MHz crystal oscillator (XO) reference [24]. The PLL integrated jitter remains below 100 fs even at near-integer channels due to low fractional spurs achieved. In the low-power mode, it achieves 96.3 fs rms jitter while consuming merely 8.2 mW, resulting in an FoM of -251.2 dB. This design is a good candidate for emerging high-throughput wireless applications, including and not limiting to 5G NR 256-QAM at mm-wave frequencies and Wi-Fi 7.

This article is organized as follows. Section II describes the phase error range reduction technique. Section III elaborates the ultra-low jitter fractional- N PLL design. Section IV discusses the dual-core VCO design. Section V presents the experimental results and Section VI draws conclusions.

II. PHASE ERROR RANGE REDUCTION TECHNIQUE FOR FRACTIONAL- N PLLS

MMD controlled by a digital DSM is commonly used for fractional- N synthesis to achieve fine frequency resolution while converting QE into colored noise instead of discrete spurious tones [18], [28]. For a first-order DSM, the accumulated phase error, Φ_e , on CKFB is up to one quantization step size, i.e., the period of the MMD input clock, T_{VCO} (see Fig. 2). The variation in Φ_e due to QE increases with the order of DSM, as there are more quantization levels in the frequency division ratio, NDIV. The amplitude of Φ_e for a second- and a third-order DSM is $2 \times T_{VCO}$ and $4 \times T_{VCO}$, respectively. Although using a first-order DSM can relax the dynamic range requirement of DTC or any other phase detectors drastically [29], a second-order or third-order DSM is usually required to provide sufficient scrambling for near-integer channels to avoid strong in-band fractional spurs to achieve sub-100 fs rms jitter [5], [23]. Consequently, the required DTC DR doubles and a more linear DTC is needed to avoid increasing phase intermodulation noise due to its nonlinearity over a larger variation in Φ_e . To achieve a linear DTC, both analog linearization techniques and digital pre-distortion were studied in literature, which demonstrated in-band fractional spur below -60 dBc at the cost of higher power consumption due to hundreds of μA bleeding current used in [16], doubled chip area using a replica branch in [21] and [22], or an additional 32-bit lookup table used in [20] for pre-distortion.

Alternatively, in this work, we use a phase error range reduction technique depicted in Fig. 3 to reduce the accumulated Φ_e on CKFB by half for a given order of DSM so that the maximum DR required in DTC is halved, reducing its integral non-linearity (INL). In addition, a smaller DR also helps reduce DTC random jitter and power consumption, which

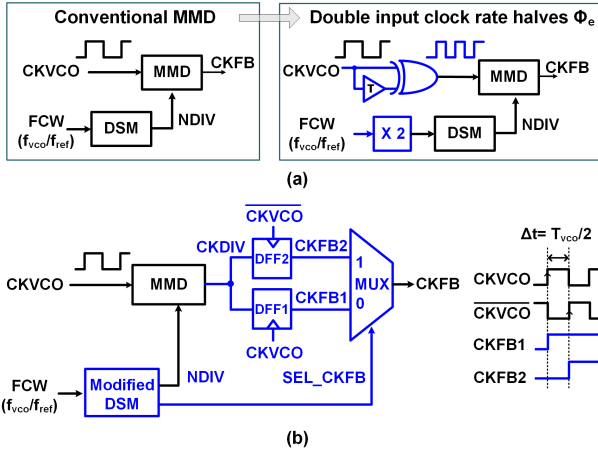


Fig. 3. (a) Reduce Φ_e by doubling MMD input clock rate. (b) Modified MMD to reduce Φ_e on CKFB by half.

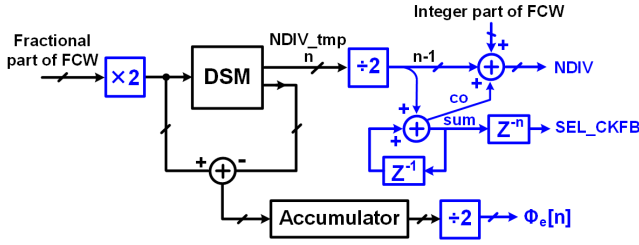


Fig. 4. Implementation of the modified digital DSM.

will be elaborated in Section III. As the amplitude of Φ_e is proportional to T_{VCO} , a brute-force way is to double the MMD input clock rate to reduce the quantization step size to $T_{VCO}/2$, as shown in Fig. 3(a). Accordingly, the frequency command word ($FCW = f_{vco}/f_{ref}$) also needs to be doubled, and then fed to the DSM to maintain the same division ratio. However, practical implementation is limited by the maximum operation frequency of MMD (e.g., <12 GHz across PVT, with 0.8-V supply voltage in a 14-nm FINFET process). Moreover, it is not easy to double the multi-gigahertz VCO clock, CKVCO, using logic gates. Alternatively, two D flip-flops (DFFs) are added after the MMD in Fig. 3(b) to retime the divide-down clock by rising and falling edges of CKVCO, respectively. Consequently, CKFB1 leads CKFB2 by $T_{VCO}/2$. The final phase of CKFB is dithered between CKFB1 and CKFB2 via a 2-to-1 mux to obtain the same benefit as doubling the MMD input frequency. A similar technique was used in [34] for a digital bang-bang PLL and in [21] for a digital SSPLL without an MMD. It can also be applied to charge-pump analog PLLs or multi-bit TDC-based digital PLLs to ease the phase detector circuit design.

Fig. 4 shows the generation circuitry for NDIV and the mux selection, SEL_CKFB. The FCW is first right shifted by 1 bit (i.e., $\times 2$), and then sent to a conventional DSM. To obtain the same feedback division ratio, the output of DSM is divided by 2 (i.e., left shift by 1 bit), which results in a fractional bit. Since an MMD can only support the integer division ratio, this fractional bit is sent to a 1-bit accumulator

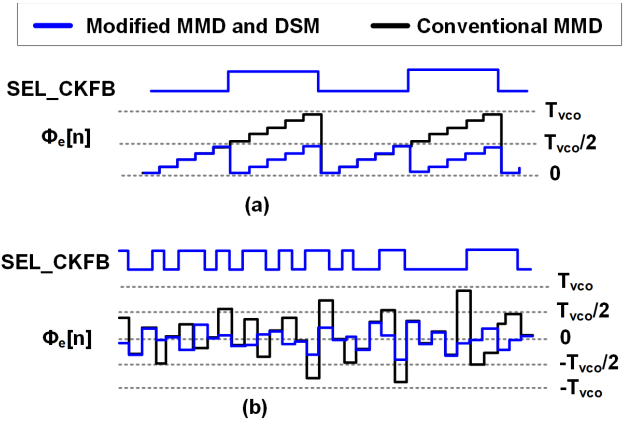


Fig. 5. Simulated Φ_e and SEL_CKFB for (a) MASH1 and (b) MASH1-1.

whose carry out, co, is added to the rest integer part to form the new NDIV. A delayed version (Z^{-n}) of the 1-bit sum is used as SEL_CKFB to account for the circuit delay in an MMD. When SEL_CKFB is “1,” an additional phase shift of $T_{VCO}/2$ is introduced on the final CKFB by selecting CKFB2. As a result, the phase quantization step size of the new MMD is reduced to $T_{VCO}/2$, which is equivalent to doubling the input clock frequency of a conventional MMD. The accumulated quantization phase error, Φ_e , can also be derived from the DSM (see Fig. 4), and then scaled by K_{DTC} to generate the DTC code in a DTC-based sampling PLL, as shown in Fig. 2. Since the proposed Φ_e range reduction is achieved by reducing the equivalent quantization step size of an MMD, its effectiveness is independent of the order and the topology of DSM. As to the MMD, it can also be any topology, e.g., 2/3 prescaler or pulse swallow. Compared with the conventional case, the achieved division ratio and fractional frequency resolution remain the same. The simulation waveforms of Φ_e and SEL_CKFB for an MASH1 and an MASH1-1 DSM are illustrated in Fig. 5(a) and (b), respectively (fractional part is 0.1). The resulted variation in Φ_e is reduced by half for both cases.

To extend this idea, quadrature phases of CKVCO could be used to further reduce the equivalent quantization step size to $T_{VCO}/4$. However, they are not readily available in most of the low-jitter PLLs, using an LC-VCO. A phase interpolator may be used to generate multiple phases, which increases the design complexity and its nonlinearity results in additional fractional spurs [30], [31].

III. DESIGN OF A LOW-JITTER SAMPLING FRACTIONAL- N PLL WITH DTC RANGE REDUCTION

A. PLL Architecture

Fig. 6 shows the block diagram of the presented low-jitter PLL. It is based on the form of a DTC-based sampling analog PLL [16]. A DTC in the reference path cancels the accumulated QE, Φ_e , on the feedback clock, CKFB. As a result, the required phase detector linear operation range after locking is reduced to a few picoseconds so that a high-gain SPD can be used. An SPD maintains the same advantages

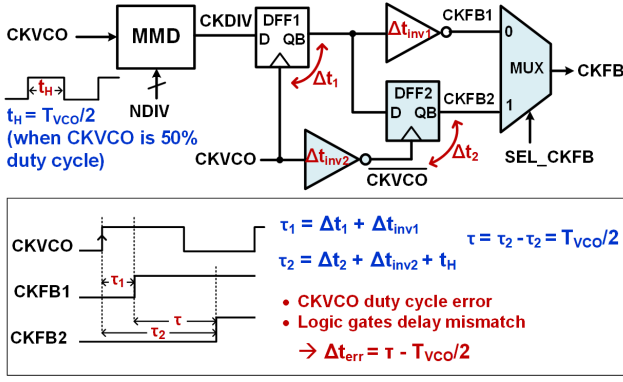


Fig. 7. Modified MMD circuit implementation.

one voltage comparator converts V_{smp} into 1-bit phase error polarity data, $e[k]$, which are then used for multiple-sign-LMS-based background calibrations, as shown in Fig. 6(a). The reference voltage of the comparator is adjusted dynamically by a first-order digital DSM to track the G_M threshold voltage so that the dc offset in $e[k]$ is removed [16], enabling K_{DTC} calibration, reference clock DCC, and VCO DCC in this analog PLL.

B. Modified MMD

In a classic MMD, the divide-down clock, CKDIV, is usually retimed by its input clock, CKVCO, (i.e., DFF1 in Fig. 7) to remove the accumulated jitter on CKFB due to asynchronous stages used for high-speed and low-power operation [32]. A half T_{VCO} delay can be introduced conceptually by adding DFF2 in parallel to sample CKDIV by inverted CKVCO, as shown in Fig. 3(b). However, it is difficult to satisfy the stringent timing requirement for both DFFs to avoid metastability. The sum of setup and hold time needs to be less than $T_{\text{VCO}}/2$ (e.g., 66 ps for CKVCO at 7.5 GHz), which is only half of the time allowed for DFF1 only case. One way to resolve this is by cascading DFF2 after DFF1, as shown in Fig. 7. The inverter delay, Δt_{inv} , is designed to be close to the clock-to-output propagation delay in DFF1 so that DFF2 has the same timing margin as DFF1 on the first order. The resultant phase relationship of CKFB1 and CKFB2 with respect to the rising edge of CKVCO is plotted in Fig. 7 as τ_1 and τ_2 , respectively, in which t_H is the time difference between the rising and falling edges of CKVCO. Consequently, the delay between CKFB1 and CKFB2 is exactly half of the VCO period in the absence of device mismatches and CKVCO duty-cycle error. As to the MMD, this work uses a classic cascaded $\div 2/3$ prescaler topology to achieve a programmable division ratio from 16 to 255 [33].

C. VCO DCC and Other Digital Calibrations

The duty-cycle error in CKVCO results in phase error, Δt_{err} , as illustrated in Fig. 7. More errors come from the inverter delay and the DFF propagation delay mismatches between the two paths. For a 5-GHz LC-VCO with a 2% duty-cycle error, and a delay mismatch of 20 ps in DFFs,

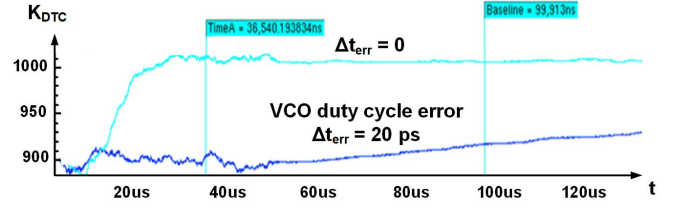
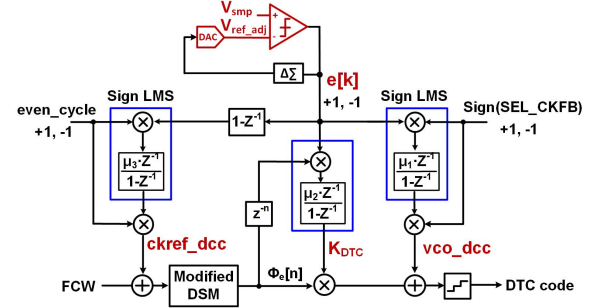
Fig. 8. Simulated K_{DTC} transient with and without VCO duty-cycle error.

Fig. 9. LMS-based digital background calibrations in the presented analog PLL.

the residue phase error is 24 ps, which is only a fraction of the 200-ps reduction achieved for a second-order DSM. Although this extra phase error is much smaller than T_{VCO} , it can be comparable to the linear operation range of an SPD, which is typically <30 ps. Moreover, it results in dc offset in the 1-bit phase error polarity data, $e[k]$, which are used for K_{DTC} calibration. In simulation, a Δt_{err} of 20 ps is sufficient to disrupt the DTC gain calibration, as shown in Fig. 8. The VCO duty-cycle error not only slows down the K_{DTC} convergence but also results in a static gain error. To resolve this, a VCO DCC loop is added in this design, which is shown in Fig. 9. A sign-extended version of SEL_CKFB is used to correlate with $e[k]$ by a sign-LMS regression. The calibration result, vco_dcc , is added to the DTC code to pull-in or push-back the phase of CKDTC accordingly to compensate the phase error due to VCO duty-cycle error at the expense of a fractional increase in DTC delay ($<10\%$).

In addition to VCO DCC, the aforementioned K_{DTC} calibration and CKREF DCC are also elaborated in Fig. 9. All these background calibrations rely on sign-LMS regression [35] and share the same phase error input, $e[k]$, from a single voltage comparator output. The other data used for correlation vary. For DTC gain calibration, it is the accumulated phase QE from DSM, $\Phi_e[n]$. For CKREF DCC, it is a bit stream alternating between $+1$ and -1 , representing even and odd cycles of CKDTC, respectively. For VCO DCC, it is SEL_CKFB as explained above. To ensure the stability of each adaptation loop and minimize the potential fighting among them, different step size, μ , is used in each LMS loop to control its convergence speed, as shown in Fig. 9. The simulated convergence transient for MASH1-1 with FCW of 45.104 is plotted in Fig. 10(a), in which VCO DCC loop converges first despite DTC gain error and CKREF duty-cycle error, both acting as interference. The CKREF DCC

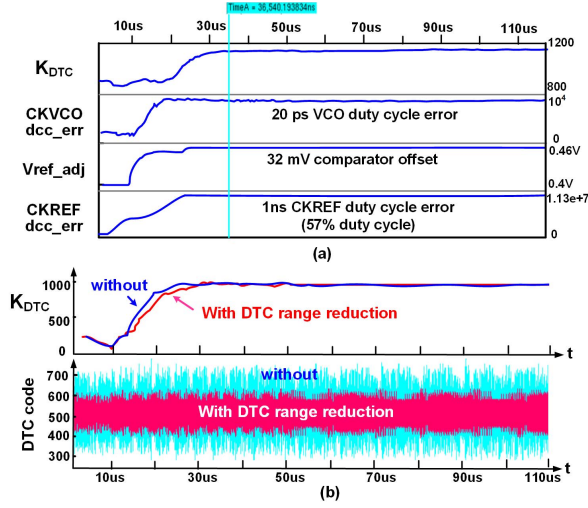


Fig. 10. (a) Digital calibration convergence transient (FCW = 45.104, MASH1-1). (b) Comparison of K_{DTC} and DTC code with and without DTC range reduction.

[i.e., CKREF dcc_err in Fig. 10(a)] converges next and is followed by K_{DTC} eventually. All the calibrations converge within 40 μ s despite large initial errors, e.g., 32-mV comparator offset, 10% K_{DTC} error, 20 ps VCO duty-cycle error, and CKREF of 57% duty cycle used in this simulation. Moreover, all three calibrations run in background simultaneously and converge robustly at both integer and near-integer channels when DSM is second-order or higher. For a first-order DSM, these calibrations have difficulty to converge in both simulation and measurement, as $\Phi_e[n]$ could exhibit very low-frequency idle tones at near-integer channels. Consequently, the LMS loop must be much slower than the period of $\Phi_e[n]$ to operate properly according to the adaptive filter theory [35]. Fig. 10(b) compares the convergence trajectories of K_{DTC} for the cases with and without DTC range reduction. Using the modified MMD and DSM reduces DTC code variation by half, without compromising background calibration speed and accuracy.

D. Digital-to-Time Converter

Fig. 11(a) shows the simplified schematic of the DTC with a clock doubler at input. The DTC consists of an RC delay core and an inverter buffer that acts as a voltage comparator. The DTC code controls the n -bit switched-capacitor bank to realize a unit delay of $T_{res} = \ln 2 \cdot R \cdot C_{LSB}$, and a total DR of $2^n \cdot T_{res}$. This topology is simple and can operate under low supply voltage (e.g., 0.8 V after supply regulation in this 14-nm design). As a contrary, a constant-slope DTC (slope = I/C) [36] may require a much higher supply voltage (e.g., 1.5 V) to minimize the current source variation over the required DR. However, in an RC delay-based DTC, the slope-dependent (i.e., code-dependent) propagation delay due to the inverter buffer, Δt_{cmp} , results in intrinsic nonlinearity, which typically limits the INL [22], as illustrated in Fig. 11(a). Thanks to the range reduction technique, the slope of V_{dly} is steeper and it also varies less across the DTC code, thus

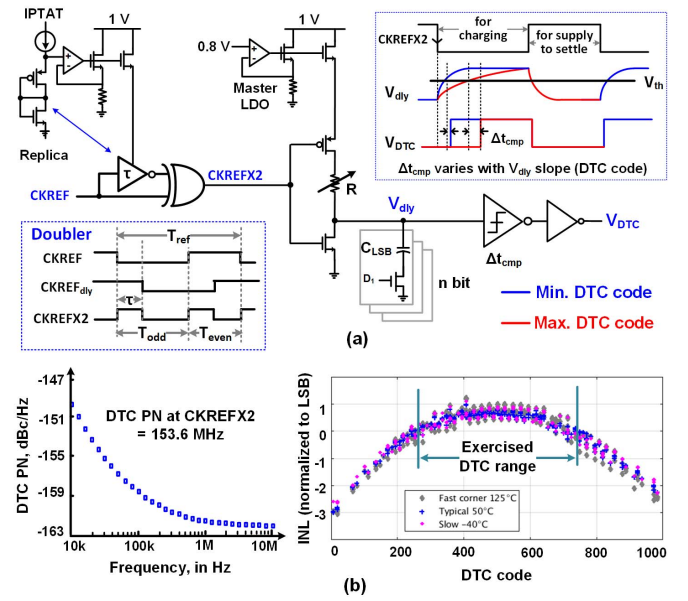


Fig. 11. (a) Schematic of the DTC with a clock double at input. (b) Simulated DTC PN and INL.

improving DTC INL. Moreover, the charging time required by the DTC core is also reduced with DR so that a higher CKDTC (i.e., f_{ref}) can be supported. In addition, a sharper slope reduces the short current consumed by both DTC core and inverter buffer, decreasing DTC power consumption. The single-side band (SSB) PN of a DTC at mid-code due to its core stage was derived in [16], and it can be rewritten to (1) as a function of DR

$$\mathcal{L} \cong 10 \cdot \log_{10} \left[2kT \cdot f_{ref}(2\pi)^2 \cdot \frac{DR^2}{\ln 2 \cdot 2^n \cdot C_{LSB}} \right]. \quad (1)$$

According to (1), DTC PN is 6 dB lower with 1/2 DR, for a given n -bit design with unit capacitor size of C_{LSB} . Besides, the sharper input slope also results in less voltage noise to jitter conversion due to the inverter buffer. Both help reduce the random jitter of CKDTC significantly.

Compared with the DTC in [16], this design reduces T_{res} from 650 to 350 fs. A 10-bit design is sufficient to cancel the QE due to a MASH1-1 DSM for a 5–7-GHz VCO. Both QN and thermal noise of the DTC are reduced by 6 dB due to the 1/2 DR, as explained above. The simulated DTC PN at 153.6-MHz sampling rate is shown in Fig. 11(b). At 1-MHz frequency offset, the DTC PN is 8 dB lower than the external XO reference clock used in this PLL (PN = -160 dBc/Hz at 1-MHz offset for the 76.8-MHz XO). The DTC PN increases at lower offset frequency due to the flicker noise from the DTC core circuit and supply regulator.

DTC INL is simulated with actual DSM sequence operating at 153.6 MHz to capture both static and dynamic distortion in the circuit. The INL curves are plotted in Fig. 12(b) for various process corners and temperature. Since only 50% of the full range is exercised with the range reduction technique, the effective INL is reduced from 4 to 1 LSB, which results in less noise folding and lower fractional spurs when used in a fractional- N PLL. To maintain INL

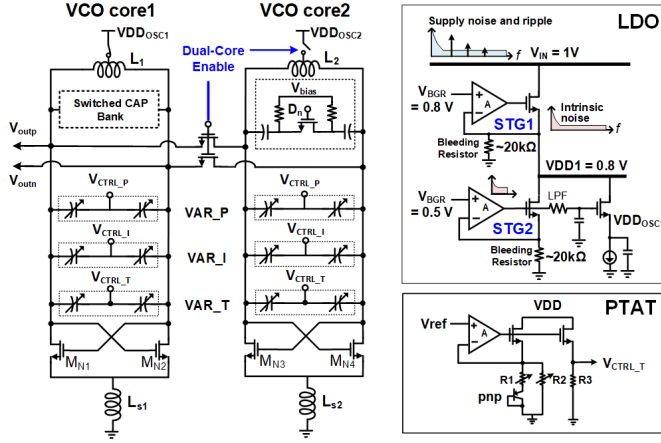


Fig. 12. Schematic of the reconfigurable dual-core VCO.

within 1 LSB over PVT variations, the programmable resistor R is adjusted to minimize the T_{res} variation over process corners. Besides, a master-slave source follower regulation scheme speeds up supply settling to within 2 ns even at slow process corner, reducing memory effect due to supply settling errors, as explained in [16]. Moreover, since a doubled clock, CKREF2X (at 153.6 MHz), is used in this DTC, its duty cycle could vary by $\pm 25\%$ over PVT due to the delay variation in the frequency doubler according to circuit simulation. In addition, CKREF also has duty-cycle error (e.g., $\pm 5\%$). These may result in insufficient time for either V_{dly} to charge to V_{th} or for the supply to settle before the next rising edge of CKREFX2 (e.g., less than 1.5 ns is available in worst case), which ultimately limits the DTC operating frequency. To resolve this, a proportional to absolute temperature (PTAT) current is fed to a replica delay stage to generate the supply voltage for the delay logic of the doubler. With this biasing scheme, the duty cycle of CKREFX2 varies $< 5\%$ over process corners and a wide temperature range (-40°C – 125°C), which is crucial for maintaining superior linearity over PVT variations at higher clock rate. Compared with [16], this work achieves similar effective INL at higher clock rate without using large bleeding current in supply regulation (0.4 mA in [16]).

IV. DUAL-CORE VCO DESIGN

Low-PN VCO is always needed despite PLL topology. Phase noise in an LC oscillator is largely determined by the inductor size and quality factor (Q). Leeson's PN equation [37] can be rewritten to show that

$$\mathcal{L} \propto \frac{f_{osc}^3 \cdot L}{V_{osc}^2 \cdot Q}. \quad (2)$$

Accordingly, for a given oscillation frequency, f_{osc} , and oscillation swing, V_{osc} (determined by the VCO topology and device reliability constrain), to reduce noise, one must scale the inductance (L) while maintaining its Q . However, there is a limit as to how small an inductor can be drawn before Q -degradation occurs, which diminishes the return on inductor scaling. To mitigate this issue, a dual-core VCO is used in this design to lower the PN by 3 dB at the cost of

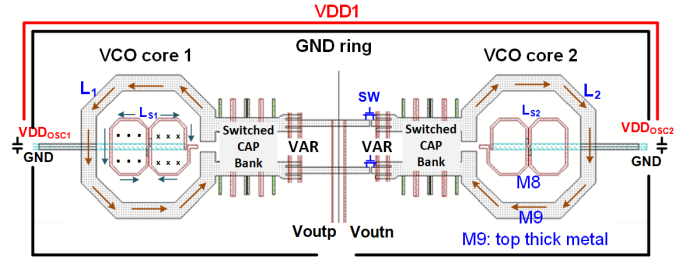


Fig. 13. Layout implementation of the dual-core VCO.

doubled current consumption, as explained in [27]. Different from the multi-core design in [25] and [26], the second core can be powered down to improve PLL power jitter FoM with negligible impact on oscillation frequency.

The complete oscillator system is shown in Fig. 13. Each core is a thin-oxide NMOS cross-coupled VCO with noise filtering inductor, L_s . An NMOS-only topology is chosen so that the VCO can operate under 1-V supply shared by the entire PLL, taking into account the voltage headroom needed for supply regulation. The tail inductor generates a high impedance around $2 \times f_{osc}$, which, in turn, prevents the differential pair transistors from conducting while in triode and degrading Q of the tank [38]. Both VCO cores are identical and are coupled together by shorting their outputs via thick-oxide NMOS switches (ON-resistance is 10Ω). Thus, the equivalent tank inductance is halved and the tank Q remains the same. The discrete tuning in each core consists of a thin-oxide switched-capacitor bank segmented into 5-binary bits and 31 thermometer bits. A 5-bit and a 3-bit bank of tunable varactors, VAR_P and VAR_I, provide continuous tuning with a programmable VCO gain for proportional ($K_{VCO,P}$) and integral path ($K_{VCO,I}$), respectively. $K_{VCO,I}$ is reconfigurable from 0 to 80 MHz/V in steps of 20 MHz/V. $K_{VCO,P}$ has a finer step size of 0.5 MHz/V to fine-tune the PLL BW for jitter optimization. Each VCO core is fed with the same control line for frequency tuning. A third bank of varactors, VAR_T, connected to a PTAT reference, provides temperature compensation, which can correct up to 2 MHz of frequency drift per degree from -40°C to $+125^\circ\text{C}$. The simulated maximum single-ended oscillation swing is $1.05 V_{pp}$, which is controlled by the VCO core supply voltage to meet the thin-oxide device reliability requirement (i.e., ≤ 1.134 V in the 14-nm FINFET process).

An ON-chip low-dropout regulator (LDO) generates 0.5 V for VCO cores from a 1-V supply voltage (see Fig. 12). It consists of two stages, STG1 and STG2, to achieve > 50 dB power supply rejection (PSR) to suppress the voltage noise and ripple from OFF-chip switching regulator and other ON-chip PLL and transceiver circuits, operating under the same supply. STG1 uses a folded-cascode error amplifier with a quiescent current of $100 \mu\text{A}$ to obtain a unity gain bandwidth of 100 MHz for better PSR and low noise. STG2 uses a master-slave topology, in which an RC low-pass filter with low cut-off frequency (e.g., 1 MHz) largely suppresses the thermal noise from the master branch. The intrinsic noise from STG1 could leak to LDO output, VDD_{OSC} , in the absence of

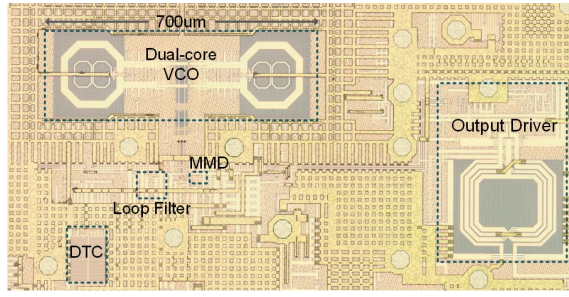


Fig. 14. Chip photomicrograph.

a closed-loop regulation in STG2. To mitigate it, the reference bandgap voltage, V_{BGR} , is pre-filtered. Besides, the bleeding resistance is reduced to 20 k Ω , and the feedback ratio is designed to be unity to reduce the thermal noise due to error amplifier and resistive feedback in STG1. Overall, the LDO achieves a PSR higher than 55 dB up to 10-MHz frequency offset, while contributing 10% of the VCO PN at 100-kHz frequency offset. Its contribution reduces to <2% at 1-MHz offset frequency.

Fig. 13 depicts the top view of the LC-tank layout including the tail inductors. The two cores are fully symmetric and star connected in the middle. Such a tank orientation minimizes the magnetic coupling between the two cores so that <1% difference in f_{osc} is observed between one-core and dual-core modes according to circuit simulation. It also prevents concurrent oscillation and ensures robust startup at the dual-core mode. For each core, the main inductor (L_1 and L_2) uses a single turn to maximize Q . The tail inductor, L_s , is folded inside the main inductor without additional chip area. It is designed as an 8-shape [39] to minimize crosstalk and Q -degradation to the main inductor. This folded layout style also facilitates placing local bypass capacitors between the local supply (i.e., center-tap of L_1) and ground, providing ac-shorts at the common-mode harmonics. This ensures that the common-mode path is localized and well-defined, which is the key to suppress flicker noise up-conversion.

V. EXPERIMENTAL RESULTS

The fractional- N PLL is implemented in a 14-nm FIN-FET process. Fig. 14 shows the chip micrograph. The PLL, excluding the output driver, occupies 0.31 mm². It synthesizes frequencies from 5 to 7 GHz from a 76.8-MHz XO reference. The complete PLL draws 14.2 mW, in which the dual-core VCO and DTC draw 12 and 1.2 mW, respectively. The MMD consumes 0.5 mW and the rest of the circuits together draw 0.5 mW, including SPD, G_M , and the digital calibration circuitries. Analog circuits share a 1-V supply and the digital circuits operate at 0.8 V. At the one-core VCO mode, the PLL power consumption is reduced to 8.2 mW.

The PLL performance is measured at the divide-by-2 test output. Fig. 15 shows the measured open-loop VCO phase noise at 3-GHz output frequency ($f_{vco} = 6$ GHz), which improves from -98.1 to -101 dBc/Hz at 100-kHz frequency

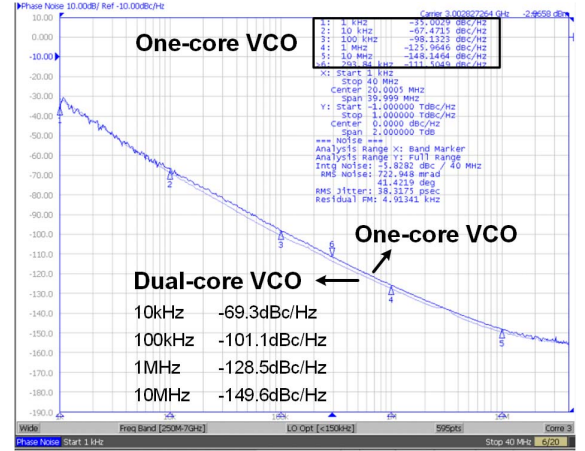
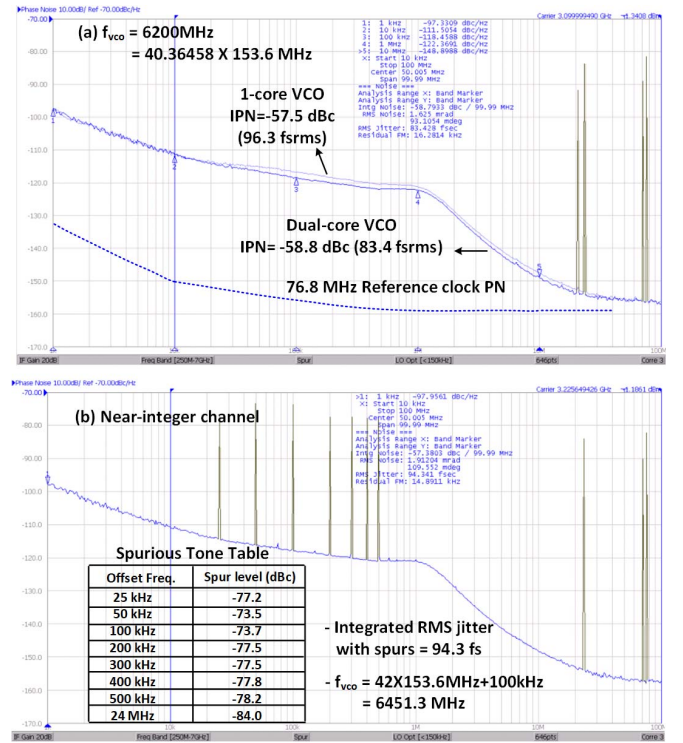


Fig. 15. Measured VCO phase noise at one-core and dual-core modes.

Fig. 16. Measured PLL phase noise at (a) $f_{vco} = 6200$ MHz = 40.36458×153.6 MHz and (b) $f_{vco} = 6451.3$ MHz = 42×153.6 MHz + 100 kHz.

offset when both VCO cores are enabled (close to the theoretical 3-dB improvement). Fig. 16 depicts the closed-loop phase noise spectrum for reference clock doubler mode. In Fig. 16(a), for a fractional channel of $f_{vco} = 6.2$ GHz, the measured integrated phase noise (IPN) at 3.1-GHz output is -58.8 dBc or 83.4 fs_{rms} with the dual-core VCO, integrated from 10 kHz to 100 MHz. The fractional spurs are located outside of the loop bandwidth, which are sufficiently low-pass filtered to contribute negligible IPN. As a contrary, Fig. 16(b) shows the PLL PN spectrum of a near-integer channel ($f_{vco} = 6451.3$ MHz = 42×153.6 MHz + 100 kHz). The rms jitter increases from 83.4 fs to 94.3 fs due to the in-band fractional spurs although they are all below -73 dBc. When

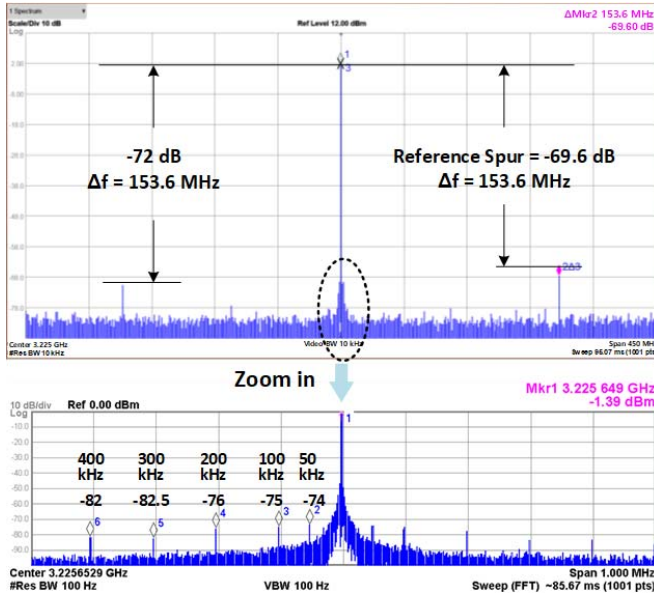


Fig. 17. Measured PLL output spectrum for $f_{vco} = 6451.3$ MHz.

configured to one-core VCO, the IPN degrades by 1.3 dB, and the PLL FoM improves from -250.1 dB (dual-core VCO) to -251.2 dB, as the PLL power consumption reduces from 14.2 to 8.2 mW. The phase noise spectrum of the external 76.8 -MHz XO is also plotted in Fig. 16(a) for reference. The PLL in-band PN is dominated by the external XO and the ON-chip reference clock buffer, which indicates that the DTC PN performance is close to the simulation results shown in Fig. 11.

The measured DTC resolution is 330 fs, reference clock duty cycle error is 180 ps, and VCO duty cycle error is 12 ps according to the calibration results. The measured worst case reference spur is -69.6 dBc from the PLL output spectrum shown in Fig. 17 for a near-integer channel of 6451.3 MHz. The measured fractional spur across fractional codes are plotted in Fig. 18(a) for MASH1-1 case, which is close to the spur level predicted by system simulation using the DTC INL curve shown in Fig. 11(b). The worst case spur is < -70 dBc without DTC nonlinearity calibration and negligible supply bleeding current.

To reveal the benefits of the DTC range reduction technique, the PLL integrated jitter including spurs is measured with and without it, as shown in Fig. 18(b). The 10-bit DTC offers a 320 -ps full range, which is not enough to support a conventional MASH1-1 DSM for f_{vco} less than 6.25 GHz. Thus, the rms jitter degrades severely when f_{vco} is less than 6 GHz. For a higher f_{vco} , although DTC range is sufficient for both cases, using the range reduction still helps decrease the integrated jitter from 120 to 90 fs due to lower fractional spurs and noise folding caused by DTC nonlinearity. The background VCO DCC loop is disabled intentionally to check its effectiveness, and the measured PLL rms jitter is also plotted in Fig. 18(b). For some channel frequencies, the DTC gain fails to converge when VCO DCC is disabled and the resulting integrated jitter is high (e.g., 180 fs). Once enabling the VCO DCC, the integrated jitter is always reduced

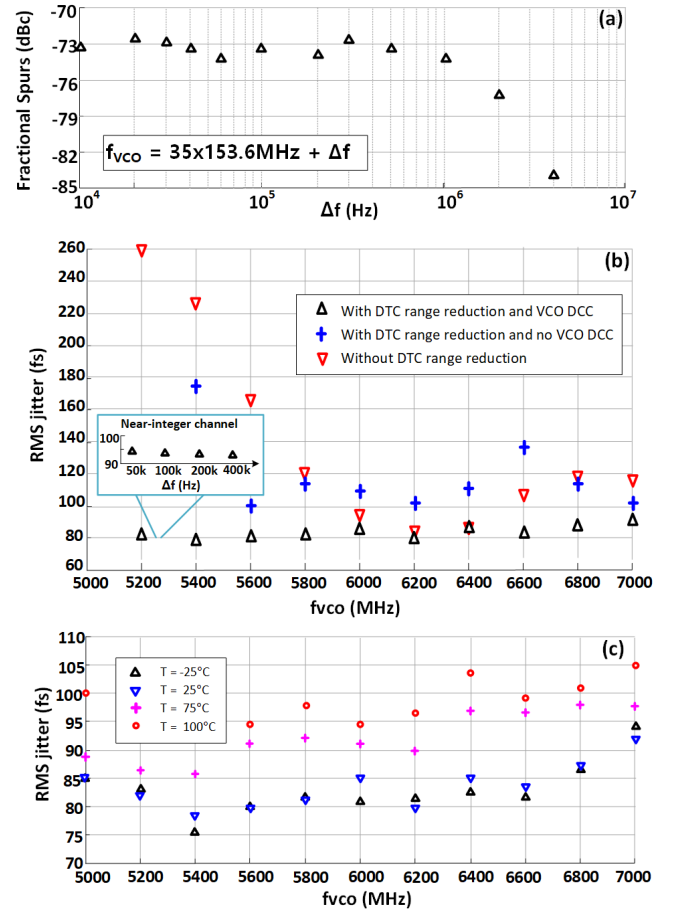


Fig. 18. (a) Worst case fractional spur measured at half of f_{vco} versus f_{vco} . (b) Measured PLL rms jitter with and without DTC range reduction across f_{vco} . (c) Measured PLL rms jitter over temperatures.

to < 100 fs including spurs, demonstrating the necessity of this background calibration in a DTC-based sampling PLL architecture, as explained in Section III.

Fig. 18(c) shows the PLL rms jitter measured from -25°C to 100°C , which degrades only 20 fs, thanks to the digital background calibrations. It also indicates good DTC linearity over temperature variation. In this test, PLL is first locked at 25°C , and then heated up or cooled down to different temperatures without relock. This demonstrates the wide temperature tracking capability due to the use of a dedicated varactor bank that is controlled by a PTAT voltage to compensate the VCO frequency drift over temperature.

Table I shows the performance summary and comparison to state-of-the-art fractional- N PLLs with external reference clock lower than 100 MHz. This PLL achieves sub- 100 fs integrated jitter, and the best power jitter FoM (< -250 dB) among them. Compared with [16], it obtains similar rms jitter with much lower power consumption, as both DTC thermal noise and QN are lower due to DTC range reduction technique so that a wider loop bandwidth (1.2 MHz versus 800 kHz in [16]) can be used to suppress VCO PN. In addition, using a higher reference clock reduces the noise multiplication factor of the reference path.

TABLE I
PERFORMANCE SUMMARY OF STATE-OF-THE-ART FRACTIONAL-*N* PLL

	This work	ISSCC'16 [19]	JSSC'17 [5]	JSSC'19 [16]	JSSC'16 [30]	ISSCC'19 [40]	ISSCC'19 [41]
Reference (MHz)	76.8X2	40	26	52X2	40	40X4 / 80X2	60
Architecture	DTC+SPD analog PLL	DTC+SPD DPLL	TDC based DPLL	DTC+SPD analog PLL	DTC+SSPD analog PLL	PFD-CP analog PLL	Triple-loop BBDPLL
Output freq. (GHz)	5 – 7	5.825	2.69	6.33	4.34 – 4.94	5.825	7 – 9
rms jitter σ_t (fs)	80–95 / 92–110* (10k to 100M)	159 (10k to 40M)	137 (10k to 10M)	75 (10k to 10M)	133.4 (10k to 10M)	110 / 82 (10k to 10M)	131 (10k to 10M)
Fractional spur (dBc)	< -72	-54	-78.6	-64	-59.2	<-60	<-70
Ref. spur (dBc)	-69.6	-78	-87.6	-70.2	-70.8	-73.4 / -66.4	<-66
Power (mW)	14.2/8.2*	8.2	13.4	18.9	6.2	22.8 / 14.7	13.4
FoM** (dB)	-250.4 – -249/ -251.6 – -250.0*	-246.8	-246	-249.7	-249.5	-245.6 / -250	-246.4
Core area (mm ²)	0.31	0.3	0.257	0.45	0.2	0.47	0.25
Technology	14 nm	28 nm	14 nm	28 nm	65 nm	28 nm	16 nm
Supply voltage (V)	1V	1.05V/1.5V	NA	1.8V/1.1V	1V	NA	0.9V

*one-core VCO mode

**FoM = $20 \cdot \log_{10}(\sigma_t / 1 \text{ s}) + 10 \cdot \log_{10}(\text{Power} / 1 \text{ mW})$

VI. CONCLUSION

This work presents a 6-GHz fractional-*N* sampling PLL with a DTC range reduction technique and a dual-core VCO in a 140 nm FINFET process. It achieves sub-100 fs integrated jitter with a 76.8-MHz crystal reference, while consuming 14.2 mW. Compared with prior work, it further reduces the performance gap between integer-*N* and fractional-*N* PLLs in terms of jitter power product and it is a good candidate for emerging high-throughput wireless applications.

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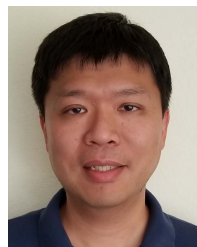
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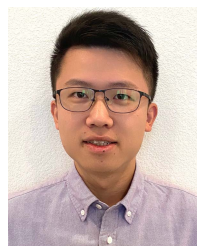
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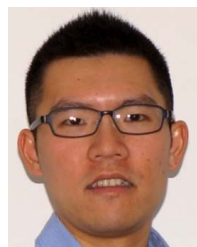
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